

REVIEW + CLARIFICATIONS

Wakes:

- $W \approx W_0 + x_w \left(W_x^D x_s - \frac{1}{2} W^Q x_w \right) + y_w \left(W_y^D y_s + W^Q y_w \right)$
- $W_{||} \triangleq \frac{\partial}{\partial z} W(\vec{\rho}_s, \vec{\rho}_w, z) \approx W'_0(z) \sim \text{consideration only for the long. problem}$
- $\vec{W}_\perp \triangleq \vec{\nabla}_\perp W = \left(\frac{\partial}{\partial x_w} W, \frac{\partial}{\partial y_w} W \right) = (W_x, W_y)$
 $\Rightarrow \vec{W}_\perp \approx \left(W_x^D x_s - W^Q x_w, W_y^D y_s + W^Q y_w \right)$
- P.W. $\rightarrow \frac{\partial}{\partial z} \vec{W}_\perp = \vec{\nabla}_\perp W_{||}$ here, $W_{||} \approx W'_0(z)$ shouldn't be used, otherwise $\vec{W}_\perp = 0$ always
- Cylind. symmetry: $W_x^D = W_y^D = W^D, W^Q = 0$
 $\Rightarrow \vec{W}_\perp = W^D (x_s, y_s) = W^D \vec{\rho}_s = W_\perp \hat{\rho}_s$

Impedances:

- $Z \approx \frac{\omega}{c} Z_{||} + x_w \left(Z_x^D x_s - \frac{1}{2} Z^Q x_w \right) + y_w \left(Z_y^D y_s + \frac{1}{2} Z^Q y_w \right)$
- Where:
 $Z \triangleq -\frac{i}{c} \int_{-\infty}^{\infty} W e^{i\omega z/c} dz$; $Z_{||} \triangleq \frac{1}{c} \int_{-\infty}^{\infty} W'_0 e^{i\omega z/c} dz$;
 $Z^{D/Q} \triangleq -\frac{i}{c} \int_{-\infty}^{\infty} W^{D/Q} e^{i\omega z/c} dz$
- $Z_{||}^{\text{full}} = \frac{1}{c} \int_{-\infty}^{\infty} \frac{\partial W}{\partial z} e^{i\omega z/c} dz \stackrel{(\#)}{=} \frac{\omega}{c} Z$
 $\Rightarrow Z_{||}^{\text{full}} = Z_{||} + \frac{\omega}{c} x_w \left(Z_x^D x_s - \dots \right) + \dots$
 $\sim Z_{||}^{\text{full}} \approx Z_{||}$

$$\frac{1}{c} \int_{-\infty}^{\infty} \frac{\partial W}{\partial z} e^{i\omega z/c} dz = \frac{1}{c} W e^{i\omega z/c} \Big|_{z=-\infty}^{z=\infty} - \frac{i\omega}{c^2} \int_{-\infty}^{\infty} W e^{i\omega z/c} dz = \dots$$

$$\Rightarrow Z_{||}^{\text{full}} = \frac{\omega}{c} \left(-\frac{i}{c} \int_{-\infty}^{\infty} W e^{i\omega z/c} dz \right) = \frac{\omega}{c} Z$$

$$\vec{Z}_\perp \triangleq \vec{\nabla}_\perp Z = \left(\frac{\partial}{\partial x_w} Z, \frac{\partial}{\partial y_w} Z \right) = (Z_x, Z_y)$$

$$\Rightarrow \vec{Z}_\perp \approx \left(Z_x^D x_s - Z^Q x_w, Z_y^D y_s + Z^Q y_w \right)$$

- P.W. $\rightarrow \vec{Z}_\perp = \vec{\nabla}_\perp Z = \frac{c}{\omega} \vec{\nabla}_\perp Z_{||}^{\text{full}}$ again, the app. $Z_{||}^{\text{full}}$ should be used only when dealing w/ the long. effect
- Cylind. symmetry:

$$Z_x^D = Z_y^D = Z^D, Z^Q = 0$$

$$\Rightarrow \vec{Z}_\perp = Z^D (x_s, y_s) = Z^D \vec{\rho}_s = Z_\perp \hat{\rho}_s$$

• Extra: Multipole expansion approach

$$w_m(r, \theta, z) = -\frac{1}{q} \text{Im} W_m(z) r^m \cos(m\theta), \quad \begin{cases} W_{||,m} = \frac{\partial}{\partial z} W_m \\ W_{\perp,m} = \vec{\nabla}_\perp W_m \end{cases}$$

$$W'_m(z) \triangleq -\frac{\partial}{\partial z} W_m(z)$$

$$Z_{||,m} \triangleq \frac{1}{c} \int_{-\infty}^{\infty} W'_m(z) e^{i\omega z/c} dz = -\frac{\omega}{c} \left(-\frac{i}{c} \int_{-\infty}^{\infty} W_m(z) e^{i\omega z/c} dz \right) = -\frac{\omega}{c} Z_{\perp,m}$$

4. The wakes can be obtained back from the impedances by the inverse transforms:

$$W_o'(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} Z_{||}(\omega) e^{-i\omega z/c} d\omega ; \quad W_t(z) = \frac{i}{2\pi} \int_{-\infty}^{\infty} Z_t(\omega) e^{-i\omega z/c} d\omega$$

where W_t refers to W_x^D, W_y^D or $W^q \sim$ same for Z_t .

5. Reality Properties:

Since the wakes are real quantities - $W_o'^* = W_o'$, $W_t^* = W_t$ - the impedances satisfy

$$Z_{||}^*(\omega) = \frac{1}{c} \int_{-\infty}^{\infty} W_o'(z) e^{-i\omega z/c} dz = Z_{||}(-\omega) \Rightarrow \begin{cases} \text{Re } Z_{||}(-\omega) = \text{Re } Z_{||}(\omega) \sim \text{even} \\ \text{Im } Z_{||}(-\omega) = -\text{Im } Z_{||}(\omega) \sim \text{odd} \end{cases}$$

$$Z_t^*(\omega) = + \frac{i}{c} \int_{-\infty}^{\infty} W_t(z) e^{-i\omega z/c} dz = -Z_t(-\omega) \Rightarrow \begin{cases} \text{Re } Z_t(-\omega) = -\text{Re } Z_t(\omega) \sim \text{odd} \\ \text{Im } Z_t(-\omega) = \text{Im } Z_t(\omega) \sim \text{even} \end{cases}$$

6. Passivity and Positivity

Using the harmonic decomposition of the beam current, the total energy lost by the beam to the chamber can be written as:

$$U_{\text{loss}} = \frac{1}{\pi} \int_0^{\infty} |\hat{I}(\omega)|^2 \text{Re} \{ Z_{||}(\omega) \} d\omega, \quad \text{where clearly } |\hat{I}(\omega)|^2 \geq 0.$$

Thus, for a passive chamber: $U_{\text{loss}} \geq 0$ for any current distribution

Since we can choose $\hat{I}(\omega)$ concentrated around any specific frequency ω $\rightarrow \text{Re} \{ Z_{||}(\omega) \} \geq 0 \quad \forall \quad \omega \geq 0$

And since $\operatorname{Re}\{Z_{||}(-\omega)\} = \operatorname{Re}\{Z_{||}(\omega)\}$, this means:

$$\operatorname{Re}\{Z_{||}(\omega)\} \geq 0 \quad \text{for all } \omega$$

Now, for the transverse impedances, something similar can be derived for the dipolar terms:

→ For a displaced beam - described by a dipole-current moment $D(\omega) \sim x_s \hat{I}(\omega)$ - the contribution of the dipole channel to the energy loss can be written:

$$U_D \propto \int_0^\infty |D(\omega)|^2 \cdot \operatorname{Re}\{Z_{||,D}(\omega)\} d\omega \quad ; \quad \text{where} \quad Z_{||,D} = \frac{1}{c} \int_{-\infty}^\infty W_{x/y}^{D'}(z) e^{i\omega z/c} dz$$

(remembering that $W_{||}^{full}(z) = \frac{\partial}{\partial z} W(z) = W'_o(z) + x_w W'_x(z) + y_w W'_y(z) + \dots$)

Integrating by parts:

$$\frac{1}{c} \int_{-\infty}^\infty W_{x/y}^{D'} e^{i\omega z/c} dz = \frac{\omega}{c} \left(-\frac{i}{c} \int_{-\infty}^\infty W_{x/y}^D e^{i\omega z/c} dz \right) = \frac{\omega}{c} Z_{x/y}^D$$

And thus, passivity implies:

$$U_D \geq 0 \quad \text{for any } D(\omega) \Rightarrow \operatorname{Re}\{Z_{||,D}(\omega)\} \geq 0 \Rightarrow \frac{\omega}{c} \operatorname{Re}\{Z_{x/y}^D\} \geq 0 \quad \forall \omega \geq 0$$

$$\therefore \text{Since } \operatorname{Re}\{Z_t(-\omega)\} = -\operatorname{Re}\{Z_t(\omega)\} : \quad \Rightarrow \operatorname{Re}\{Z_{x/y}^D\} \geq 0 \quad \forall \omega \geq 0$$

$$\begin{cases} \operatorname{Re}\{Z_{x/y}^D(\omega)\} \geq 0, & \omega \geq 0 \\ \operatorname{Re}\{Z_{x/y}^D(\omega)\} \leq 0, & \omega < 0 \end{cases}$$

* For the quadrupolar impedance, passivity does not imply $\operatorname{Re}\{Z^Q(\omega)\} \geq 0$ because, while

$$U_D \propto |D|^2 \frac{\omega}{c} \operatorname{Re}\{Z_{x/y}^D(\omega)\} \quad \left(\text{which} \Rightarrow \omega \cdot \operatorname{Re}\{Z^D\} \geq 0 \right), \quad U_Q \propto \operatorname{Re}\left\{ Z^Q \overset{\text{monopole moment}}{I_o} \overset{\text{quad. moment}}{Q_b^*} \right\} \cdot \frac{\omega}{c}, \quad \text{and } I_o Q_b^* \text{ can have any sign}$$

7. Causality and Kramers - Kronig

Due to the causality of the wakes - $W(z < 0) = 0$ -, the real and imaginary parts of the impedances cannot be completely arbitrary: they satisfy the so-called Kramers - Kronig relations, i.e., are connected by a Hilbert transform.

To derive and outline such connexion, let's write $Z_{||}(\omega) = R_{||}(\omega) + iX_{||}(\omega)$.

Then, since the wakes are real:

$$\begin{cases} R_{||}(\omega) = \text{Re}\{Z_{||}(\omega)\} \Rightarrow R_{||}(-\omega) = R_{||}(\omega) \\ X_{||}(\omega) = \text{Im}\{Z_{||}(\omega)\} \Rightarrow X_{||}(-\omega) = -X_{||}(\omega) \end{cases}$$

From the inverse transform:

$$\begin{aligned} W_{||}(z) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} Z_{||}(\omega) e^{-i\omega z/c} d\omega = \frac{1}{2\pi} \left[\int_{-\infty}^0 Z_{||}(\omega) e^{-i\omega z/c} d\omega + \int_0^{\infty} Z_{||}(\omega) e^{-i\omega z/c} d\omega \right] = \\ &= \frac{1}{2\pi} \left[\int_0^{\infty} (R_{||}(-\omega) + iX_{||}(-\omega)) e^{-i\omega z/c} d\omega + \int_0^{\infty} (R_{||}(\omega) + iX_{||}(\omega)) e^{i\omega z/c} d\omega \right] = \\ &= \frac{1}{2\pi} \int_0^{\infty} \left[R_{||}(\omega) \cdot \overbrace{(e^{i\omega z/c} + e^{-i\omega z/c})}^{2 \cos(\omega z/c)} + iX_{||}(\omega) \cdot \overbrace{(e^{i\omega z/c} - e^{-i\omega z/c})}^{-2i \sin(\omega z/c)} \right] d\omega = \\ &= \frac{1}{\pi} \int_0^{\infty} [R_{||}(\omega) \cos(\omega z/c) + X_{||}(\omega) \sin(\omega z/c)] d\omega \end{aligned}$$

Now, to make use of causality, let $u > 0$ and evaluate $w_{II}(z = -u)$:

$$w_{II}(-u) = \frac{1}{\pi} \int_0^{\infty} \left[R_{II}(\omega) \cos\left(\frac{\omega u}{c}\right) - X_{II}(\omega) \sin\left(\frac{\omega u}{c}\right) \right] d\omega = 0$$

$$\Rightarrow \int_0^{\infty} R_{II}(\omega) \cos\left(\frac{\omega u}{c}\right) d\omega = \int_0^{\infty} X_{II} \sin\left(\frac{\omega u}{c}\right) d\omega$$

Then, for $z > 0$, we can write (from (I)):

$$w_{II}(z) = \frac{2}{\pi} \int_0^{\infty} R_{II}(\omega) \cos\left(\frac{\omega z}{c}\right) d\omega \quad (\text{II}) \quad \text{or} \quad w(z) = \frac{2}{\pi} \int_0^{\infty} X_{II}(\omega) \sin\left(\frac{\omega z}{c}\right) d\omega \quad (\text{III})$$

$\Rightarrow *$ The knowledge of either $\text{Re}\{Z_{II}\}$ or $\text{Im}\{Z_{II}\}$ alone is enough to reconstruct the wake w_{II}

Now, to derive Kramers-Kronig relations we see that:

$$Z_{II}(\omega) = \frac{1}{c} \int_{-\infty}^{\infty} w_{II}(z) e^{i\omega z/c} dz = \frac{1}{c} \int_0^{\infty} \left[w_{II} \cos\left(\frac{\omega z}{c}\right) + i w_{II} \sin\left(\frac{\omega z}{c}\right) \right] dz = R_{II}(\omega) + i X_{II}(\omega)$$

$$\Rightarrow R_{II}(\omega) = \frac{1}{c} \int_0^{\infty} w_{II}(z) \cos\left(\frac{\omega z}{c}\right) dz \stackrel{(\text{III})}{=} \frac{2}{\pi c} \int_0^{\infty} \int_0^{\infty} X_{II}(\omega') \sin\left(\frac{\omega' z}{c}\right) \cos\left(\frac{\omega z}{c}\right) d\omega' dz =$$

$$= \frac{2}{\pi c} \int_0^{\infty} \left(\int_0^{\infty} \frac{1}{2} \left[\sin\left(\frac{(\omega' + \omega)z}{c}\right) + \sin\left(\frac{(\omega' - \omega)z}{c}\right) \right] dz \right) d\omega'$$

$$\begin{aligned} \sin(a+b) &= \sin a \cos b + \sin b \cos a \\ \sin(a-b) &= \sin a \cos b - \sin b \cos a \\ \Rightarrow 2 \sin a \cos b &= \sin(a+b) + \sin(a-b) \end{aligned}$$

Using the strategy of introducing a small damping factor $e^{-\eta z}$ ($\eta > 0$)

$$\int_0^{\infty} \sin\left(\frac{az}{c}\right) dz = \lim_{\eta \rightarrow 0^+} \int_0^{\infty} e^{-\eta z} \sin\left(\frac{az}{c}\right) dz = \lim_{\eta \rightarrow 0^+} \frac{ac}{a^2 + \eta^2 c^2} = \frac{c}{a}$$

However, if "a" can pass through zero inside another, like in our case of $a = \omega' - \omega$, this result must be interpreted in the ^{Cauchy} principal value sense: $c \cdot P \frac{1}{a}$.

$$\frac{1}{\omega' + \omega} + \frac{1}{\omega' - \omega} = \frac{2\omega'}{\omega'^2 - \omega^2}$$

Thus:

$$R_{11}(\omega) = \frac{1}{\pi c} \int_0^{\infty} \left(\frac{c}{\omega' + \omega} + P \frac{c}{\omega' - \omega} \right) X_{11}(\omega') d\omega' = \frac{2}{\pi} P \int_0^{\infty} \frac{\omega' X_{11}(\omega')}{\omega'^2 - \omega^2} d\omega'$$

And analogously:

$$X_{11}(\omega) = \frac{1}{c} \int_0^{\infty} \omega_{11}(z) \sin\left(\frac{\omega z}{c}\right) dz \quad \xrightarrow{[II] + algebra} \quad X_{11}(\omega) = -\frac{2\omega}{\pi} P \int_0^{\infty} \frac{R_{11}(\omega')}{\omega'^2 - \omega^2} d\omega'$$

Or equivalently, opening $\frac{2\omega'}{\omega'^2 - \omega^2}$ back into $\frac{1}{\omega' + \omega} + \frac{1}{\omega' - \omega}$ and manipulating to express these relations in terms of the whole frequency axis, gives the standard Kramers - Kronig relations:

$$R_{11}(\omega) = \frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{X_{11}(\omega')}{\omega' - \omega} d\omega'; \quad X_{11}(\omega) = -\frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{R_{11}(\omega')}{\omega' - \omega} d\omega'$$

Proceeding analogously for the transverse impedances ($Z_t(\omega) = R_t(\omega) + i X_t(\omega)$) also gives:

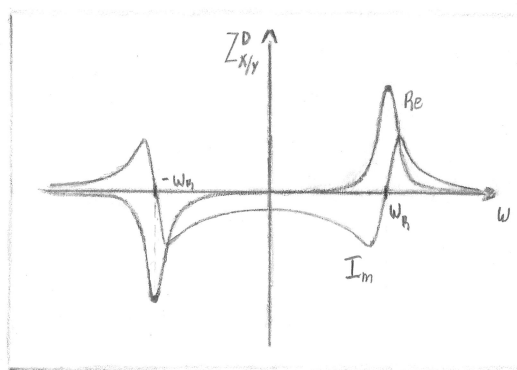
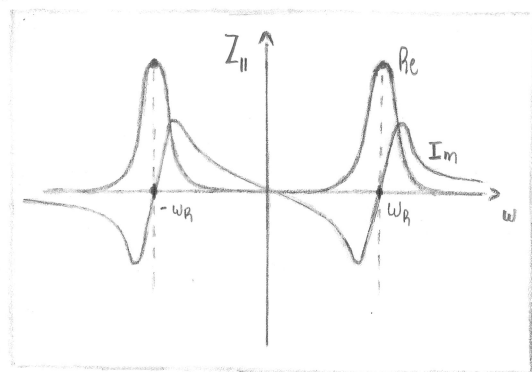
$$R_t(\omega) = \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{X_t(\omega')}{\omega' - \omega} d\omega'$$

$$X_t(\omega) = -\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{R_t(\omega')}{\omega' - \omega} d\omega'$$

~ Kramers-Kronig relations for the transverse impedances

To illustrate many of the presented properties of the real and imaginary

parts of $Z_{||}(\omega)$ and $Z_t(\omega)$, we can draw these curves for the case of a resonator model - one of the models that will be explored in the next section - with resonant frequency ω_R :



Adapted from [S. Khan]